

Adsorbent	Adsorbate	Surface area	Adsorption Initial Conc	RE (%)	Contact Time	Dose(g/L)	RPM	Initial pH	
Graphene	Cr3+	136	45	1000	90.65	120	1.3	125	4
Zero-valent	Cr3+	117.97	35.3	25	70	90	1	125	3
Copolymer	Cr3+	2630	82.4	50	93	45	1	125	1.1
Graphene	Cr3+	157	2859.38	14	93	90	10	125	1.5
Graphene	Cr3+	6.53	92.65	52	92.65	12	1	125	5
Modified g	Cr3+	1.52	21.57	75	98.2	60	400	125	2
Acrylonitril	Cr3+	7.13	101.2	30	80	420	0.6	125	2
Syzygium j	Cr3+	2.49	35.3	110	100	240	5	125	2
Green alga	Cr3+	4.33	61.485	50	98	40	2	125	1
Jatropha w	Cr3+	8.71	123.6	65	90.65	360	1.3	125	6
Tamarind v	Pb2+	2.49	35.3	30	28	40	2	125	6.5
Pterocladia	Cr3+	4.65	66	50	100	120	6.5	125	1
Zizania cad	Cr3+	0.19	2.7	30	84.8	48	0.8	125	2.5
Prawn shell	Cr3+	7.04	100	75	98	31.4	1	125	5
Iron oxide	Cr3+	4.49	63.69	175	71	120	2	125	2
Ozone-treated	Cr3+	0.77	10.9	75	86	150	4	125	2
Coconut cc	Cr3+	2.11	29.96	30, 0.1	98	120	1	125	8
Coconut cc	Cr3+	5.37	76.3	40	96	90	3.2	125	2
Acinetobacter	Cr3+	2.49	35.3	55	94	2880	1	125	7
Succinic acid	Cr3+	2.49	35.3	50	92	60	20	125	2
Sugarcane	Cr3+	2.49	35.3	77.5	93	60	6.85	125	8.58
Wheat bran	Cr3+	6.55	93	20	89	20	80	125	5
Wheat bran	Cr3+	9.37	133	75	99.9	300	2	125	1.5
Wheat bran	Cr3+	0.37	5.28	200	90	730	20	125	2.1
Modified c	Cr3+	2.73	38.77	60	94	60	50	125	2
Coconut cc	Cr3+	1.95	27.7	250	70	7200	1	125	3
Modified o	Cr3+	0.28	4	250	82	4320	4	125	4
Pre-treated	Cr3+	0.66	9.43	10	65	260	10	125	3
Composite	Cr3+		5.37	10	71	60	5		1
Hydrochloric	Cr3+	0.12	1.7	50	84	360	60	125	3
Sulphuric acid	Cr3+	1.43	20.3	45	90.65	240	4	125	2
Egg shell m	Cr3+	11.27	160	100	90	725	25	125	3
Silica functionalized	Cd2+	0.16	1	25	80	2160	20	125	5.3
Amino functionalized	Cd2+	2.88	18.25	50	90	120	5	125	5
Iminodiacetic	Cd2+		36.4	103	98.8	1440	4	125	5.6
Polyamine	Cd2+	5.58	35.3	5.62	70	2880	1	125	6.25
α-Ketoglutaric	Cd2+	31.8	201.2	300	93	90	0.04	125	6
Nano-hydroxy	Cd2+	16.91	107	100	92	90	5	125	5.6
Polyaniline	Cd2+	19.28	122	130	99.6	120	4.5	125	6
O-carboxymethyl	Cd2+	5.58	35.3	675	95	1440	1	125	10
Multi-walled	Cd2+	5.58	35.3	50	94	120	1	125	6.5
Synthetic zeolite	Cd2+	71.63	453.25	500	90.65	180	1	125	5
Zeolite	Cd2+	655		25	60	90	25	125	6
Zeolite from	Cd2+	19.91	126	2248.2	95.6	1440	10	125	6.6
Oil shale in	Cd2+	15.11	95.6	100	90.65	750	1	125	7
Natural zeolite	Cd2+	32	9	50	71	1440	10	125	5

Coffee resi Cd2+	2.8	17.7	130	90	30	10	125	2
Coffee hus Cd2+	1.09	6.9	75	75	4320	6.7	125	4
Coffee bea Cd2+	106.25	672.3	505.85	90	1440	10	125	8
Coffee groi Cd2+	2.47	15.65	350	87	120	9	125	7
Rice husk ε Cd2+	1.98	12.5	50	29.8	5	5.5	125	6
Sulphuric α Cd2+	6.31	39.955	100	90.65	60	1	100	7
Olive cake Cd2+	10.34	65.4	100	66	1440	0.3	100	6
Zinc chloric Cd2+	0.11	0.68	45	95	90	30	125	6.15
Microwave Cd2+	1.85	11.72	20	95.32	7	1.1	200	5
Activated c Cd2+	9.02	57.098	2,000.00	23	200	6	400	5
Olive stone Cd2+	1.42	8.97	3372.3	49.2	60	13.33	30	5.8
Succinic an Cd2+	7.6	48.095	45	80	95	20.4	140	4
Xanthate n Cd2+	17.78	112.51	65	99.7	30	4.1	125	4
Activated c Cd2+	14.76	93.4	25	98	120	0.3921	125	5
Green cocc Cd2+	1.55	9.78	20	98	120	5	175	7
Green cocc Cd2+	3.9	24.67	100	90.65	5.66	1.62	125	5
Aspergillus Cu2+	2.01	13.4	10	93.3	120	1	200	6
Rhizopus o Cu2+	2.91	19.4	110	19.4	200	1	150	5
Pleurotus ꞑ Cu2+	0.58	3.86	100	90	12	3	150	4
Trametes v Cu2+	9.15	60.98	60	65.38	80	1	150	5.51
Penicillium Cu2+	4.45	29.68	50	76.2	30	1.5	120	5
Saccharom Cu2+	0.39	2.59	110	43.08	120	15	125	3.5
Padina sp. Cu2+	7.63	50.87	127	90	30	2	200	5
Cystoseira Cu2+	24.01	160	40	100	720	2.5	190	4.5
Sargassum Cu2+	15.32	102.1	50	87.3	225	0.1	150	4.5
Codium ve Cu2+	2.54	16.9	80	90.65	120	0.5	5000	5
Spirogyra i Cu2+	2.9	19.063,19.	80	90.65	120	5	5000	4
Spirulina p Cu2+	14.21	94.7	250	91	120	2.5	150	6.7
Dried micr Cu2+	3.68	24.5	100	90	120	0.4	125	4
Paenibacill Cu2+	128.61	857	260	99	120	1	125	6
Pseudomo Cu2+	5.43	36.2	300	20	30	1	5000	5
Pseudomo Cu2+	0.99	6.6	0.1	80	10	1	5500	5
Sphaerotil Cu2+	9	60	100	99	150	3	125	6
Bacillus sp. Cu2+	1.34	8.945	100	17.89	30	2	125	5
Sawdust Cu2+	630	37.17	100	90	60	10	150	6.6
Granulatec Ni2+	0.44	2.88	50	28.8	120	0.05	125	4.5
Montmoril Ni2+	699	12.89	100	100	360	1	200	6
Original be Pb2+	109.8	19.19	15	78	10	0.1	5000	5
Cicer arien Pb2+	4.49	27.79	100	82	1440	0.05	200	4
Pine wood Pb2+	2.73	4.13	100	100	1440	10	12	5
AC F-400 Pb2+	984	30.11	100	100	1440	10	12	5
CNTs Pb2+	2.82	17.44	20	83	240	0.05	125	7
Activated (Cd2+	1000	178.5	30	86	20	0.02	100	2
Activated (Pb2+	1000	294.1	50	74	20	0.02	100	2
Activated (Cr3+	1000	60.9	100	29.6	20	0.02	100	2
Activated (Ni2+	1000	400	200	84.1	20	0.02	100	2
Activated (Zn2+	1000	344.8	30	83.6	20	0.02	100	2

CaO/Fe3O4-Cr3+	36.47	6.4	5	98.7	50	2	400	5
bentonite/Cd2+	52.18	35	5	92	20	1	200	6
CaO/Fe3O4-Co2+	71.23	217.39	5	94.32	30	2	400	6
Cobalt-ferr Cd2+	17.84	303	20	100	50	20	125	6
Fe3O4/talcu Cu2+	37.079	74.6	100	72.15	2	0.12	200	5
Watermel Cu2+	15.55	5.73	10	98	20	0.5	125	5
Watermel Cu2+	4.69	31.25	10	88	10	1	4000	5
Watermel Cu2+	0.94	6.281	10	58.4	600	0.5	125	5
Watermel Cu2+	5.88	39.2	300	98.1	120	2.5	125	5
Watermel Cu2+	1.44	9.57	10	80	120	1	125	8
Watermel Zn2+	3.47	22.55	275	52.4	120	1.5	125	6
Watermel Zn2+	5.43	35.3	1	90.3	60	0.02	125	7
Watermel Ni2+	2.83	18.43	100	97	120	5	125	6.69
Watermel Ni2+	5.98	38.98	60	69	120	0.1	150	6
Watermel Ni2+	5.42	35.3	100	70	30	2.5	150	5
Watermel Pb2+	8.88	55	500	72	15	1	125	5
Watermel Pb2+	18.77	116.2	300	98.1	120	2.5	125	5
Watermel Pb2+	5.7	35.3	5	77.3	60	0.02	125	6
Watermel Pb2+	15.84	98.06	10	99.9	600	0.5	125	6.8
Watermel Cd2+	10	63.29	125	35	30	1.5	150	5
Watermel Tl+	41.75	178.4	10.25	98	30	10	200	11
Watermel Tl+	262.8	1123	10.25	98.5	30	2	600	9
Watermel As3+	52.36	100	4	99	120	1	4000	8.2
Watermel As5+	5.02	94	5	98	120	1	4000	4.6
Watermel Fe2+	1	4.98	2.2	98.3	45	0.04	125	7
Watermel Mn2+	0.32	1.37	0.7	98.9	45	0.05	125	6
Watermel Co2+	1.94	23.3	50	57	30	2	150	6
Montmorill Pb2+	4.62	28.62	50	99	50	1	200	5
Fe3O4-Chit Cr6+	4.37	62.1	60	90	120	0.05	200	2
GO/Fe3O4 Pb2+	20.45	126.6	5	94	50	3	200	6.2
MSCMNPs-Hg2+	31.56	276.56	50	66.1	20	50	200	4
CNSR-Fe3(Cd2+	0.86	5.46	100	95	50	2	200	8
DNPH/SDS Cr6+	11.94	169.5	50	96	90	0.05	200	4
NPSFe3O4 Cr6+	3.1	44	50	90	50	1	200	5
glycol-Fe3(Cd2+	5.7	35.3	100	92	10	1	200	6
SDS-nzVI Cr6+	25.83	366.7	50	97.25	30	1	200	5
Geothite, FeCu2+	17.54	116.855	50	98.177	120	0.03	200	5.2
As-receiver Cr6+	0.15	2.18	20	96	1440	24	150	6
CSC coated Cr6+	0.26	3.65	20	71	1440	18	150	4
H2SO4-treated Cr6+	0.29	4.05	20	93.4	1440	15	150	6
HNO3-treated Cr6+	0.77	10.88	20	99.6	1440	12	150	4
Cocoa shell Cd2+	0.78	4.94	28.1	26.5	1440	5	500	2
Cocoa shell Cr3+	0.18	2.52	13	99.1	1440	5	500	2
Cocoa shell Cu2+	0.43	2.87	15.88	58	1440	5	500	2
Cocoa shell Ni2+	0.4	2.63	14.68	99.253	1440	5	500	2
Cocoa shell Zn2+	0.45	2.92	16.35	95	1440	5	500	2
Pecan shell Cu2+	14.26	95	500	96.954	30	1	125	3.6

Partheniun Ni2+	8.34	54.35	70	97.5	720	1	300	5
Orange pei Ni2+	0.92	6.01	25	88	1440	1	180	7
Orange pei Cu2+	0.55	3.65	25	99	1440	1	180	7
Orange pei Zn2+	0.81	5.25	25	100	1440	1	180	7
Sawdust Cr6+	1.11	15.82	100	99	120	2.4	125	2
Maple saw Cr6+	0.36	5.1	10	90.3	5	50	3000	5
PSA Zn2+	2.14	13.9	32.7	99	1440	10	500	4.8
Polyamine Cd	39.98	35.3, 253,1	100	70	2880	1	180	6.25
Î±-Ketoglut Cd	30.88	201.2	1,000	93	90	0.04	180	6
Electrospun Cd	38.08	248.1	525	72	120	1	180	5
Nano-hydr Cd	14.54	92	300	92	90	5	180	5.6
Polyaniline Cd	22.25	145	130,40	99.6	120	4.5	180	6
O-carboxyl Cd	5.58	35.3	674.46	95	1440	1	125	10
Copolymer Cr	12.65	82.4	50	93	45	1	160	1.1
Graphene Cr	438.83	2859.38	14	93, 90	90	10, 0.12	125	1.5
Modified g Cr	3.31	21.57	50	98.2	60	400	150	2
Acrylonitril Cr	15.53	101.2	30	80	420	0.6	125	2
Syzygium j. Cr	2.49	35.3	60	100	240	5	125	2
Green alga Cr	17.24	112.36	125	98	40	2	125	1
Tamarind v Cr	1.97	28.019	30	28	40,60	2	125, 120	6.5,5.68
Pterocladia Cr	10.13	66	40	100	120	6.5	200	1
Zizania cad Cr	0.41	2.7	30	84.8	48	0.8	125	2.5
Prawn shel Cr	15.44	100.6	75	98	31.4	1	160	5.5
Composite Cr	40.59	264.5	100	95	240	2.5	125	2
Iron oxide Cr	9.77	63.69	175	71	120	2	125	2
Modified ri Cr	2.98	42.378	190, 25	95, 91.48	5760, 90	8.5	125	6.8, 2
Coconut cc Cr	4.6	29.96	30	98	120	1	125	8
Coconut cc Cr	11.71	76.3	60	96	90	3.25	200	2
Acinetobac Cr	2.49	35.3	55	90, 19	2880	1	125	7
Succinic ac Cr	2.49	35.3	50	92	60	20	125	2
Sugarcane Cr	2.49	35.3	77.5	87	60	6.85	125	8.58
Wheat bra Cr	20.41	133	50	99.9	300	2	125	1.5
Wheat bra Cr	0.81	5.28	52	90	440	20	125	2.2
Modified c Cr	11.71	76.3	60	90	60	50	125	2
Coconut cc Cr	10.8	70.4	255	70	7200	1	125	3
Modified o Cr	6.22	40.56	250	82	4320	4	125	4
Pre-treat Cr	0.34	4.79	10	51	260	10	125	3
Composite Cr	0.13	1.9	30	71	60	5	125	1
Hydrochlor Cr	0.26	1.7	50	84	360	60	125	3
A. Hypoge Cr6+	0.66	4.3	20	90.65	60	1	125	5
Almond Cr6+	1.57	10.2	510	81	120	1	125	2
Banana pei Cd2+	5.23	34.1	450	90.65	60	1	150	5.5
Banana pei cu2+	8.04	52.4	450	91	60	1	150	5.5
Banana pei Pb2+	3.97	25.9	450	89	60	1	150	5.5
Banana pei Ni2+	1.2	54.4	450	85	60	1	150	5.5
Graphene Pb2+	39.29	256	3.225	99	120	1	125	5
Graphene Cd2+	21.02	136.98	3.225	98.46	120	1	125	6

Graphene Cr	14.22	92.65	52	92.65	12	1	125	5.5
Graphene Cu2+	31.81	207.27	50	86.77	300	1	125	5
Graphene As3+	6.56	42.75	5	80.39	900	1	125	5
Graphene Co2+	3.27	21.28	13.5	93.8	30	1	125	5.5
Graphene Ni2+	10.13	66.01	55	78.31	120	1	125	6.5
Graphene Hg2+	43.09	280.8	50	98	120	1	125	5
Graphene Zn2+	31.97	208.33	10	100	1440	1	125	7
Activated c Pb2+	20.6	134.22	40	97.95	50	1	125	6.5
Activated c Cd2+	5.84	38.03	200	95	120	1	125	6.8
Activated c Cr	3.31	21.57	75	98.2	60	1	125	5
Activated c Cu2+	5.83	38	37	97	75	1	125	5
Activated c As5+	2.15	14	40	99	150	1	125	6.5
Activated c Co2+	3.46	22.57	100	90.3	100	1	125	6
Activated c Ni2+	10.37	67.56	20	94	120	1	125	5.8
Activated c Hg2+	23.63	154	40	100	40	1	125	5
Activated c Zn2+	4.77	31.11	200	95	120	1	125	4.5
Activated c Mn2+	7.86	51.23	20	88	120	1	125	5.8
Activated c Fe2+	9.02	58.76	20	90	120	1	125	5.8
Rice husk Cd	19.33	125.94	210	99.1	120	1	125	6.5
Rice husk Cr	9.77	63.69	175	71	120	1	125	2
Rice husk Fe2+	1.3	0.22	11.78	99.253	135	1	125	6
Rice husk Cu2+	1.3	0.1	5.43	98.177	135	1	125	6
Rice husk Ni2+	1.3	0.029	1.74	96.954	135	1	125	6
Zeolite Pb2+	2.58	16.81	100	92	25	1	125	5
Zeolite Cd2+	7.21	47	2500	97.5	30	1	125	4
Zeolite Cr	0.5	3.27	2.89	96	240	1	125	3.5
Zeolite Cu2++	0.32	2.1	250	99	60	1	125	5
Zeolite Co2+	1.7	11.1	75	66.1	60	1	125	3.5
Zeolite Ni2+	3.78	24.65	20	99	60	1	125	7.5
Zeolite Hg2+	2.51	16.354	294	90	420	1	125	5.5
Zeolite Zn2+	4.39	28.6	20	99	30	1	125	7.5
LMS Pb2+	9.6	33.9	50	85	180	2	250	6
AFL Pb2+	14.03	91.4	200	80	180	1	300	6
M-Lignin-P Pb2+	1.1	96.66	50	99.73	60	4	150	6
MAL Pb2+	0.49	79.9	20	97	100	1	150	6
GAL Pb2+	18.28	119.1	200	96	120	2	200	6
DML Pb2+	3.13	208.3	30	100	120	0.6	150	5
L-CNTs Pb2+	130.75	235	10	98.3	50	0.5	150	5.8
Lignin-base Pb2+	2.14	250	30	98.66	240	0.8	150	5
PAN/SL CN Pb2+	174.885	588.24	100	79	240	40	200	5
P(AANa-co Pb2+	17	630	400	93.7	60	0.4	180	5
MLBB Pb2+	186	952	1000	98	300	0.6	150	7
LS-GH Pb2+	459.3	1259	400	90	30	1	4000	5
Lignin-base Pb2+	0.8109	365.43	210	98	120	0.5	150	5
Surface fur Pb2+	22.3	188	20	99	30	0.01	100	5
MoS2 @LC Cr6+	14	198.7 Å 20.0		99.35	10	0.1	9000	2
LGNs Cr6+	25.98	368.78	200	90	1440	1	300	2

LGH	Cr6+	66.9067	1743.9	300	88	1080	0.33	100	2
Lignin/PVA	Cr6+	24.71	350.87	250	60	1440	0.1	180	2
Fluorescen	Cr6+	42.25	599.9	120	90	160	0.5	150	7
PEI-lignin	Cr6+	46.34	657.9	200	85.44	1440	1	180	2
lignin-PEI	Cr6+	20.32	898.2	1300	80.48	2880	1	150	2
LSMA	Cr6+	89.09	1264.8	50	95.4	60	0.33	150	2
Lignin-base	Cr6+	749.16	887.8	100	100	720	0.5	150	2
Magnetic	Cr3+	91	44.56	100	90	360	0.5	150	5
Carbon-lig	Cr3+	30.24	382	100	64.1	180	0.2	150	7
AL-SH	Hg	11.72	125	200	90	480	1	150	4
PSD-LS	Hg	25.28	164.7	175	80	120	1	150	6
PEI-CS-L	Hg	101.83	663.5	664.1	83.5	1	1	150	5.5
N-MPC-ED	Cd	1130	43.68	10	84	45	0.025	150	6
BSB	Cd	14.8	61.8	100	99	60	1	300	6
LBA	Cd	13.41	87.4	200	95	180	1	350	6
Corncob-b	Cd	3.935	85.75	100	95	120	1	180	4
BLAC	Ni	2943	14.025	20	76.6	480	1	150	6.39
SLS-based	Ni	44.97	293	500	93	120	0.5	150	6
Lignin-Silic	Co	92	18	3.5	63	90	1.14	180	7
Lignin-PNN	Ag	17.27	1556.8	2000	86.4	2880	2	180	5.65
Zn- lignin	As	255	20.2	100	80.2	60	2	4000	6
Activated I	As	9.59	62.5	20	100	60	0.5	150	4
Lignin-base	V	8	10.79	55	45	120	0.3	800	5
FALA	Cu	0.42	64.14	4.91	97.78	100	0.8	150	6
FALA	Pb	0.42	65.65	4.42	99.82	100	0.8	150	6
CFL	Cu	10.57	68.9	150	74	120	2	150	6
CFL	Pb	9.02	58.8	150	71	120	1.5	150	6
Mg2Al-LS-I	Cu	0.7	64	50	47	10	0.5	10000	5.7
Mg2Al-LS-I	Pb	0.7	123	50	72.8	10	0.5	10000	5.7
SLCA	Cu	42.36	276	100	82	60	0.2	150	7
SLCA	Pb	49.57	323	100	85	60	0.2	150	7
PG/L/SA	Cd	131.4	79.88	50	95	30	0.25	150	6
PG/L/SA	Pb	131.4	226.24	100	85	30	0.25	150	6
LPC	Cd	375.8	126.37	60	92	60	0.4	150	6
LPC	Pb	375.8	250.47	60	96	60	0.4	150	6
CSL	Cu	43.43	283	100	99	60	0.5	150	6
CSL	Co	59.24	386	100	99	60	0.5	150	6
GG/SLS	Cu	108.81	709	500	81	180	0.2	150	7
GG/SLS	Co	92.24	601	500	79	180	0.2	150	7
CHL1	Ni	2.7	5.28	30	88	60	5	1000	7
CHL2	Cd	2.7	5.9	30	98.4	60	5	1000	7
PLCD	Cu	20.6	98	20	94.3	20	0.4	150	6
PLCD	Zn	20.6	78	20	76.5	20	0.4	150	6
PLCD	Ni	20.6	67	20	65.1	20	0.4	150	6
chitin/ligni	Ni	1.86	70.41	50	85.1	60	5	1000	5
chitin/ligni	Cu	1.86	75.7	50	85.5	60	5	1000	5
chitin/ligni	Zn	1.86	82.41	50	92.3	60	5	1000	5

chitin/ligni Pb	1.86	91.74	50	99.1	60	5	1000	5
PAA-g-APL Pb	1.5198	222.73	207.2	68	780	1	150	5
PAA-g-APL Cu	1.5198	20.69	63.5	28.6	780	1	150	5
PAA-g-APL Cd	1.5198	6.61	112.4	3.4	780	1	150	5
A-LMS-5 Cr	7.68	54.2	0.18	95	180	0.5	150	6
A-LMS-6 As	7.68	53.12	0.22	96	180	0.5	150	6
A-LMS-7 Cd	7.68	74.84	0.15	97	180	0.5	150	6.5
A-LMS-8 Ni	7.68	49.42	0.19	89	180	0.5	150	6.5
Sulfo-meth Co	22.27	145.14	200	91.2	250	2	150	7
Sulfo-meth Cu	18.05	117.64	200	88.5	250	2	150	7
Sulfo-meth Ni	25.59	166.72	200	80.1	250	2	150	7
Sulfo-meth Cd	42.65	277.89	200	97.7	250	2	150	7
Sulfo-meth Pb	52.92	344.85	200	99.1	60	2	150	7
Peanut she As3+	28.11	0.16	1.005	79.6	2160	5	150	5.5
Buckwheat As3+	11.4	0.48	0.8025	36.44	2160	5	150	5.5
Buckwheat As3+	11.4	0.48	0.6	37.5	2160	5	150	5.5
Buckwheat As3+	11.4	0.48	1.005	29.1	2160	5	150	5.5
Buckwheat As3+	10.72	0.54	0.8025	32.71	2160	5	150	5.5
Buckwheat As3+	10.72	0.54	0.6	36.25	2160	5	150	5.5
Buckwheat As3+	10.72	0.54	1.005	34.32	2160	5	150	5.5
Buckwheat As3+	17.02	0.458	1.005	35.07	2160	5	150	5.5
Buckwheat As3+	17.02	0.458	0.8025	29.9	2160	5	150	5.5
Buckwheat As3+	17.02	0.458	0.6	41.25	2160	5	150	5.5
Buckwheat As3+	17.8	0.254	0.6	35	2160	5	150	5.5
Buckwheat As3+	17.8	0.254	1.005	20.14	2160	5	150	5.5
Buckwheat As3+	17.8	0.254	0.8025	31.77	2160	5	150	5.5
Corncoobs(C As3+	12.44	0.495	0.6	33.75	2160	5	150	5.5
Corncoobs(C As3+	12.44	0.495	1.005	26.11	2160	5	150	5.5
Corncoobs(C As3+	12.44	0.495	0.8025	32.7	2160	5	150	5.5
Corncoobs(C As3+	14.42	0.553	0.6	33.75	2160	5	150	5.5
Corncoobs(C As3+	14.42	0.553	0.8025	29.9	2160	5	150	5.5
Corncoobs(C As3+	37.11	0.06	0.6	11.25	2160	5	150	5.5
Corncoobs(C As3+	37.11	0.06	1.005	5.97	2160	5	150	5.5
Corncoobs(C As3+	37.11	0.06	0.8025	7.47	2160	5	150	5.5
Corncoobs(C As3+	47.63	0.32	0.8025	22.42	2160	5	150	5.5
Corncoobs(C As3+	47.63	0.32	0.6	23.75	2160	5	150	5.5
Corncoobs(C As3+	47.63	0.32	1.005	21.64	2160	5	150	5.5
Mulberry v As3+	58.03	0.08	1.005	5.97	2160	5	150	5.5
Mulberry v As3+	58.03	0.08	0.6	8.75	2160	5	150	5.5
Mulberry v As3+	58.03	0.08	0.8025	7.47	2160	5	150	5.5
Mulberry v As3+	16.56	0.03	0.8025	3.73	2160	5	150	5.5
Mulberry v As3+	16.56	0.03	1.005	2.23	2160	5	150	5.5
Mulberry v As3+	16.56	0.03	0.6	3.75	2160	5	150	5.5
Mulberry v As3+	24.46	0.036	0.6	6.25	2160	5	150	5.5
Mulberry v As3+	24.46	0.036	1.005	3.73	2160	5	150	5.5
Mulberry v As3+	24.46	0.036	0.8025	2.8	2160	5	150	5.5
Peanut she As3+	18.58	0.23	0.8025	22.42	2160	5	150	5.5

Peanut she As3+	18.58	0.23	1.005	20.14	2160	5	150	5.5
Peanut she As3+	18.58	0.23	0.6	21.25	2160	5	150	5.5
Peanut she As3+	18.58	0.23	0.6	26.25	2160	5	150	5.5
Peanut she As3+	28.11	0.14	0.8025	16.82	2160	5	150	5.5
Peanut she As3+	14.03	0.29	1.005	23.88	2160	5	150	5.5
Peanut she As3+	14.03	0.29	0.6	23.75	2160	5	150	5.5
Peanut she As3+	14.03	0.29	0.8025	23.36	2160	5	150	5.5
Peanut she As3+	14.08	0.18	0.6	28.75	2160	5	150	5.5
Peanut she As3+	14.08	0.18	0.8025	17.75	2160	5	150	5.5
Canna indi Cd2+	3.46	73.17	200	35.99933	1440	1	180	6
Canna indi Cd2+	7.73	96.38	200	48.19	1440	1	180	7
Canna indi Cd2+	7.73	44.19	200	22.09	1440	1	180	3
Canna indi Cd2+	10.4	149.52	200	74.84825	1440	1	180	7
Canna indi Cd2+	7.73	90.26	200	45.13	1440	1	180	4
Canna indi Cd2+	3.46	74.72	200	37.34825	1440	1	180	5
Canna indi Cd2+	7.73	97.96	200	48.98	1440	1	180	5
Canna indi Cd2+	3.46	31.95	200	15.9004	1440	1	180	3
Canna indi Cd2+	3.46	81.76	200	40.64748	1440	1	180	8
Canna indi Cd2+	5.81	109.02	200	54.35027	1440	1	180	8
Canna indi Cd2+	10.4	84.95	200	42.55283	1440	1	180	3
Canna indi Cd2+	5.81	98.5	200	42.47	1440	1	180	5
Canna indi Cd2+	10.4	146.45	200	72.84735	1440	1	180	4
Canna indi Cd2+	3.46	71.14	200	35.44852	1440	1	180	7
Canna indi Cd2+	5.81	98.08	200	48.80283	1440	1	180	6
Canna indi Cd2+	5.81	91.93	200	45.25067	1440	1	180	4
Canna indi Cd2+	5.81	97.29	200	48.34757	1440	1	180	7
Canna indi Cd2+	7.73	168.8	200	84.89771	1440	1	180	6
Canna indi Cd2+	5.81	44.81	200	22.39771	1440	1	180	3
Canna indi Cd2+	10.4	150.76	200	75.55081	1440	1	180	6
Canna indi Cd2+	10.4	153.39	200	76.89973	1440	1	180	5
Canna indi Cd2+	7.73	176.79	200	89.00067	1440	1	180	8
Canna indi Cd2+	3.46	64.31	200	32.45279	1440	1	180	4
Canna indi Cd2+	10.4	159.85	200	80.05283	1440	1	180	8
Vetiveria zi Cd2+	48.99	5.84	10.0036	58.4	1440	1	180	7
Phragmite: Cd2+	5.901	17.47	49.99552	34.94	1440	1	180	7
Zizania cad Cd2+	84.05	14.8	10.0036	58.2	1440	1	180	7
Vetiveria zi Cd2+	48.99	17.17	99.99104	17.17	1440	1	180	7
Phragmite: Cd2+	5.901	29.35	79.99508	36.68	1440	1	180	7
ThaliaDeal Cd2+	21.9	43.23	79.99508	54.03	1440	1	180	7
Zizania cad Cd2+	84.05	23.47	99.99104	23.47	1440	1	180	7
ThaliaDeal Cd2+	21.9	46.64	99.99104	46.64	1440	1	180	7
Vetiveria zi Cd2+	48.99	11.53	49.99552	23.06	1440	1	180	7
Zizania cad Cd2+	84.05	8.94	19.99596	44.7	1440	1	180	7
Zizania cad Cd2+	84.05	18.7	79.99508	23.37	1440	1	180	7
Phragmite: Cd2+	5.901	34.94	99.99104	34.94	1440	1	180	7
ThaliaDeal Cd2+	21.9	39	49.99552	78	1440	1	180	7
Vetiveria zi Cd2+	48.99	15.76	79.99508	19.7	1440	1	180	7

Rice straw Cd2+	12.78	28.58	49.99552	57.16	720	1	150	6
Rice straw Cd2+	24.61	89.38	100.0023	89.38	720	1	150	6
Rice straw Cd2+	12.78	75.76	149.9978	51.84	720	1	150	6
Rice straw Cd2+	24.61	86.45	100.0023	86.45	720	1	150	8
Rice straw Cd2+	12.78	55.71	100.0023	55.71	720	1	150	4
Rice straw Cd2+	41.12	41.75	49.99552	83.5	720	1	150	6
Rice straw Cd2+	12.78	51.59	100.0023	51.59	720	1	150	5
Rice straw Cd2+	12.78	12.79	100.0023	14.37563	720	1	150	2
Rice straw Cd2+	41.12	8.35	10.0036	83.4699	720	1	150	6
Rice straw Cd2+	24.61	21.08	29.99956	70.27	720	1	150	6
Rice straw Cd2+	41.12	75.76	100.0023	75.76	720	1	150	6
Rice straw Cd2+	12.78	44.97	100.0023	44.97	720	1	150	3
Rice straw Cd2+	41.12	92.46	179.9974	51.37	720	1	150	6
Rice straw Cd2+	41.12	62.73	79.99508	78.41	720	1	150	6
Rice straw Cd2+	24.61	64.35	100.0023	64.35	720	1	150	4
Rice straw Cd2+	41.12	77.73	100.0023	77.73	720	1	150	8
Rice straw Cd2+	24.61	113.24	179.9974	62.91	720	1	150	6
Rice straw Cd2+	24.61	13.96	100.0023	13.96	720	1	150	2
Rice straw Cd2+	12.78	77.53	200.0046	38.76	720	1	150	6
Rice straw Cd2+	12.78	15.78	29.99956	52.6	720	1	150	6
Rice straw Cd2+	24.61	134.7	200.0046	67.35	720	1	150	6
Rice straw Cd2+	24.61	67.09	100.0023	67.09	720	1	150	5
Rice straw Cd2+	41.12	59.43	100.0023	59.43	720	1	150	3
Rice straw Cd2+	41.12	77.71	100.0023	77.71	720	1	150	7
Rice straw Cd2+	24.61	9.25	10.0036	92.5	720	1	150	6
Rice straw Cd2+	12.78	11.95	19.99596	59.75	720	1	150	6
Rice straw Cd2+	41.12	82.129	119.9982	68.44	720	1	150	6
Rice straw Cd2+	41.12	74.68	100.0023	74.68	720	1	150	4
Rice straw Cd2+	12.78	77.87	179.9974	43.26	720	1	150	6
Rice straw Cd2+	24.61	15.346	19.9959	76.73	720	1	150	6
Rice straw Cd2+	41.12	93.28	200.0046	46.64	720	1	150	6
Rice straw Cd2+	24.61	67.13	100.0023	67.13	720	1	150	7
Rice straw Cd2+	24.61	108.09	149.9978	72	720	1	150	6
Rice straw Cd2+	12.78	64.73	119.9982	53.33	720	1	150	6
Rice straw Cd2+	12.78	55.91	100.0023	55.91	720	1	150	8
Rice straw Cd2+	24.61	94.34	119.9982	78.33	720	1	150	6
Rice straw Cd2+	41.12	22.115	29.99956	73.33	720	1	150	6
Rice straw Cd2+	24.61	66.81	79.99508	83.51	720	1	150	6
Rice straw Cd2+	41.12	10.99	100.0023	11	720	1	150	2
Rice straw Cd2+	24.61	53.73	100.0023	53.73	720	1	150	3
Rice straw Cd2+	12.78	56.9	100.0023	56.9	720	1	150	6
Rice straw Cd2+	24.61	43.113	49.99552	86.22	720	1	150	6
Rice straw Cd2+	41.12	88.43	149.9978	58.95	720	1	150	6
Rice straw Cd2+	41.12	13.08	19.99596	65.4	720	1	150	6
Rice straw Cd2+	12.78	53.83	100.0023	53.83	720	1	150	7
Rice straw Cd2+	41.12	76	100.0023	76	720	1	150	5
Rice straw Cd2+	12.78	7.44	10.0036	74.37	720	1	150	6

Rice straw(Cd2+	12.78	45.79	79.99508	57.23	720	1	150	6
Plum stone€Cd2+	443	0.43	10.0036	43.82	2880	10	250	8
Grape stall Cd2+	72	5.16	28.57208	99.29	1440	5.5	250	5
Grape husl Cd2+	77	2.79	15.38756	100	1440	5.5	250	5
Wheat stra Cd2+	364	3.58	19.99596	98.59	1440	5.5	250	5
Nut shells (Cd2+	465	0.93	10.0036	55.95	1920	6	250	6.5
Rice husk(f Ni2+	20.1	0.129	58.69	12.9	1440	5	200	7
wheat stra Ni2+	23.2	0.3312	19.56	99.36	1440	5	200	5
Rice husk(f Ni2+	20.1	0.1275	58.69	12.75	1440	5	200	6
wheat stra Ni2+	26.4	0.0485	58.69	4.85	1440	5	200	2
wheat stra Ni2+	23.2	0.2492	14.67	99.68	1440	5	200	5
wheat stra Ni2+	26.4	0.9966	58.69	95.34	1440	5	200	10
Rice husk(f Ni2+	42	0.1238	7.33	99.04	1440	5	200	5
wheat stra Ni2+	23.2	0.1997	11.73	99.85	1440	5	200	5
Rice husk(f Ni2+	42	0.0608	58.69	6.08	1440	5	200	2
wheat stra Ni2+	23.2	0.3298	29.34	65.96	1440	5	200	5
wheat stra Ni2+	26.4	0.1905	19.56	57.15	1440	5	200	5
Rice husk(f Ni2+	20.1	0.0874	5.869	87.4	1440	5	200	5
wheat stra Ni2+	26.4	0.1808	14.67	72.32	1440	5	200	5
Rice husk(f Ni2+	42	0.1797	29.34	35.94	1440	5	200	5
wheat stra Ni2+	26.4	0.1238	7.33	99.04	1440	5	200	5
Rice husk(f Ni2+	42	0.1863	58.69	18.63	1440	5	200	5
wheat stra Ni2+	23.2	0.1658	9.78	99.46	1440	5	200	5
Rice husk(f Ni2+	42	0.1034	58.69	10.34	1440	5	200	3
wheat stra Ni2+	26.4	0.249	58.69	24.9	1440	5	200	7
wheat stra Ni2+	26.4	0.1247	58.69	12.47	1440	5	200	3
Rice husk(f Ni2+	42	0.2313	58.69	23.13	1440	5	200	8
Rice husk(f Ni2+	20.1	0.091	6.52	81.9	1440	5	200	5
wheat stra Ni2+	23.2	0.3912	58.69	39.12	1440	5	200	5
wheat stra Ni2+	26.4	0.997	58.69	99.7	1440	5	200	9
Rice husk(f Ni2+	20.1	0.0986	9.78	59.14	1440	5	200	5
Rice husk(f Ni2+	42	0.1638	14.67	65.52	1440	5	200	5
Rice husk(f Ni2+	42	0.0997	5.86	99.7	1440	5	200	5
Rice husk(f Ni2+	20.1	0.1211	58.69	12.11	1440	5	200	4
Rice husk(f Ni2+	20.1	0.9983	58.69	99.83	1440	5	200	10
Rice husk(f Ni2+	42	0.1104	6.52	99.36	1440	5	200	5
wheat stra Ni2+	23.2	0.0931	58.69	9.31	1440	5	200	2
Rice husk(f Ni2+	20.1	0.9798	58.69	97.98	1440	5	200	9
wheat stra Ni2+	26.4	0.1592	9.78	95.5	1440	5	200	5
wheat stra Ni2+	26.4	0.2673	58.69	26.73	1440	5	200	8
Rice husk(f Ni2+	20.1	0.0805	58.69	8.05	1440	5	200	3
Rice husk(f Ni2+	20.1	0.1168	29.34	23.36	1440	5	200	5
wheat stra Ni2+	23.2	0.3887	58.69	38.87	1440	5	200	7
Rice husk(f Ni2+	42	0.9983	58.69	99.83	1440	5	200	9
wheat stra Ni2+	23.2	0.3876	58.69	38.76	1440	5	200	4
Rice husk(f Ni2+	20.1	0.0964	8.38	67.45	1440	5	200	5
Rice husk(f Ni2+	42	0.1716	19.56	51.48	1440	5	200	5

Rice husk(f Ni2+	20.1	0.1016	11.73	50.8	1440	5	200	5
wheat stra Ni2+	23.2	0.248	58.69	24.8	1440	5	200	3
Rice husk(f Ni2+	20.1	0.105	14.67	42	1440	5	200	5
wheat stra Ni2+	26.4	0.2366	58.69	23.66	1440	5	200	4
Rice husk(f Ni2+	42	0.1972	58.69	19.72	1440	5	200	7
Rice husk(f Ni2+	20.1	0.094	7.33	75.2	1440	5	200	5
wheat stra Ni2+	23.2	0.3953	58.69	39.53	1440	5	200	8
Rice husk(f Ni2+	42	0.1375	8.38	96.22	1440	5	200	5
Rice husk(f Ni2+	42	0.1972	58.69	19.72	1440	5	200	6
wheat stra Ni2+	26.4	0.2164	29.34	43.28	1440	5	200	5
wheat stra Ni2+	26.4	0.2528	58.69	25.28	1440	5	200	6
Rice husk(f Ni2+	42	0.1582	11.73	79.1	1440	5	200	5
Rice husk(f Ni2+	20.1	0.1577	58.69	15.77	1440	5	200	8
Rice husk(f Ni2+	42	0.1835	58.69	18.35	1440	5	200	4
Rice husk(f Ni2+	20.1	0.1089	19.56	32.67	1440	5	200	5
wheat stra Ni2+	23.2	1	58.69	100	1440	5	200	10
Rice husk(f Ni2+	42	0.1484	9.78	89.02	1440	5	200	5
Rice husk(f Ni2+	20.1	0.0603	58.69	6.03	1440	5	200	2
wheat stra Ni2+	26.4	0.2385	58.69	23.85	1440	5	200	5
wheat stra Ni2+	23.2	0.3889	58.69	38.89	1440	5	200	6
wheat stra Ni2+	23.2	0.9985	58.69	99.85	1440	5	200	9
Rice husk(f Ni2+	20.1	0.1242	58.69	12.42	1440	5	200	5
wheat stra Ni2+	26.4	0.1401	8.38	98.04	1440	5	200	5
wheat stra Ni2+	26.4	0.1717	11.73	85.85	1440	5	200	5
Rice Husk (Pb2+	198.11	1.2	400	14.83	1440	50	160	5
Wheat str̄ Pb2+	14.37	1.64	150	54.89	1440	50	160	5
Rice Husk (Pb2+	198.11	0.55	50	55.94	1440	50	160	5
Wheat str̄ Pb2+	14.37	0.915	50	91.5	1440	50	160	5
Wheat str̄ Pb2+	14.37	2.459	400	30.73	1440	50	160	5
Rice Husk (Pb2+	198.11	1.095	300	18.25	1440	50	160	5
Rice Husk (Pb2+	198.11	0.96	200	24.01	1440	50	160	5
Wheat str̄ Pb2+	14.37	2.701	500	27.01	1440	50	160	5
Wheat str̄ Pb2+	14.37	1.568	100	78.42	1440	50	160	5
Rice Husk (Pb2+	198.11	0.901	150	30.045	1440	50	160	5
Rice Husk (Pb2+	198.11	0.73	100	36.73	1440	50	160	5
Wheat str̄ Pb2+	14.37	2.17	300	36.24	1440	50	160	5
Rice Husk (Pb2+	198.11	1.26	500	12.6	1440	50	160	5
Wheat str̄ Pb2+	14.37	1.84	200	46.13	1440	50	160	5
Grape stall Pb2+	72	11.33	62.49	99.8	1470	5.5	250	5
Grape husl Pb2+	77	17.43	99.99	95.89	1470	5.5	250	5
Nut shells (Pb2+	465	2.44	13.15	99.68	1470	5.5	250	5
Wheat str̄ Pb2+	364	18.1	99.99	100	1470	5.5	250	5
Plum stone Pb2+	443	2.267	12.49	99.83	1470	5.5	250	5
Buckwheat Pb2+	11.4	1.89	0.99456	100	2160	5	150	5.5
Buckwheat Pb2+	11.4	1.89	0.80808	97.43	2160	5	150	5.5
Buckwheat Pb2+	11.4	1.89	0.60088	100	2160	5	150	5.5
Buckwheat Pb2+	10.72	1.375	0.80808	97.43	2160	5	150	5.5

Buckwheat Pb2+	10.72	1.375	0.60088	100	2160	5	150	5.5
Buckwheat Pb2+	10.72	1.375	0.99456	100	2160	5	150	5.5
Buckwheat Pb2+	17.02	1.92	0.80808	100	2160	5	150	5.5
Buckwheat Pb2+	17.02	1.92	0.99456	100	2160	5	150	5.5
Buckwheat Pb2+	17.02	1.92	0.60088	100	2160	5	150	5.5
Buckwheat Pb2+	17.8	1.34	0.60088	100	2160	5	150	5.5
Buckwheat Pb2+	17.8	1.34	0.99456	100	2160	5	150	5.5
Buckwheat Pb2+	17.8	1.34	0.80808	100	2160	5	150	5.5
Corncobs(C Pb2+	12.44	1.05	0.60088	100	2160	5	150	5.5
Corncobs(C Pb2+	12.44	1.05	0.80808	97.43	2160	5	150	5.5
Corncobs(C Pb2+	12.44	1.05	0.99456	100	2160	5	150	5.5
Corncobs(C Pb2+	14.42	1.59	0.60088	100	2160	5	150	5.5
Corncobs(C Pb2+	14.42	1.59	0.80808	97.43	2160	5	150	5.5
Corncobs(C Pb2+	14.42	1.59	0.99456	100	2160	5	150	5.5
Corncobs(C Pb2+	37.11	1.37	0.60088	96.55	2160	5	150	5.5
Corncobs(C Pb2+	37.11	1.37	0.99456	95.83	2160	5	150	5.5
Corncobs(C Pb2+	37.11	1.37	0.80808	97.43	2160	5	150	5.5
Corncobs(C Pb2+	47.63	1.52	0.99456	97.91	2160	5	150	5.5
Corncobs(C Pb2+	47.63	1.52	0.60088	100	2160	5	150	5.5
Corncobs(C Pb2+	47.63	1.52	0.80808	97.43	2160	5	150	5.5
Mulberry v Pb2+	58.03	2.47	0.80808	100	2160	5	150	5.5
Mulberry v Pb2+	58.03	2.47	0.60088	100	2160	5	150	5.5
Mulberry v Pb2+	58.03	2.47	0.99456	100	2160	5	150	5.5
Mulberry v Pb2+	16.56	2.39	0.99456	100	2160	5	150	5.5
Mulberry v Pb2+	16.56	2.39	0.80808	97.43	2160	5	150	5.5
Mulberry v Pb2+	16.56	2.39	0.60088	100	2160	5	150	5.5
Mulberry v Pb2+	31.45	2.38	0.60088	100	2160	5	150	5.5
Mulberry v Pb2+	31.45	2.38	0.99456	100	2160	5	150	5.5
Mulberry v Pb2+	31.45	2.38	0.80808	97.43	2160	5	150	5.5
Mulberry v Pb2+	24.46	1.96	0.99456	100	2160	5	150	5.5
Mulberry v Pb2+	24.46	1.96	0.80808	100	2160	5	150	5.5
Mulberry v Pb2+	24.46	1.96	0.60088	100	2160	5	150	5.5
Peanut she Pb2+	18.58	1.21	0.60088	100	2160	5	150	5.5
Peanut she Pb2+	18.58	1.21	0.80808	97.43	2160	5	150	5.5
Peanut she Pb2+	18.58	1.21	0.99456	100	2160	5	150	5.5
Peanut she Pb2+	28.11	1.69	0.80808	100	2160	5	150	5.5
Peanut she Pb2+	28.11	1.69	0.60088	100	2160	5	150	5.5
Peanut she Pb2+	28.11	1.69	0.99456	100	2160	5	150	5.5
Peanut she Pb2+	14.03	1.32	0.80808	97.43	2160	5	150	5.5
Peanut she Pb2+	14.03	1.32	0.60088	100	2160	5	150	5.5
Peanut she Pb2+	14.03	1.32	0.99456	100	2160	5	150	5.5
Peanut she Pb2+	14.08	1.93	0.99456	100	2160	5	150	5.5
Peanut she Pb2+	14.08	1.93	0.60088	100	2160	5	150	5.5
Peanut she Pb2+	14.08	1.93	0.80808	97.43	2160	5	150	5.5
Pine wood Zn2+	219.35	0.345	6.36	8.125	1440	25	120	3
Jarrah woc Zn2+	309.29	1.32	4.7	18.25	1440	25	120	3
Pine wood Zn2+	219.35	0.59	2.03	17.125	1440	25	120	4

Jarrah woc Zn2+	309.29	31.85	19.46	38.625	1440	25	120	5
Jarrah woc Zn2+	309.29	2.15	69.58	36.125	1440	25	120	4
Pine wood Zn2+	219.35	0.78	46.29	20.25	1440	25	120	5

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